Ryan Tabrizi

rtabrizi@berkeley.edu | (650) 804-6468 | ryantabrizi.com | github.com/rtabrizi

EDUCATION

University of California, Berkeley May, 2025
B.S. Electrical Engineering & Computer Science GPA: 3.754

- **Relevant Courses:** Graduate NLP (A+), Graduate CV, Computational Photography and Vision, Intro to ML (A+), Deep Learning for CV, Convex Optimization, Discrete Math and Probability Theory, Operating Systems, Data Structures, Stanford CS231n (audit)
- **Honors and Organizations:** EECS Honors Program (Cognitive Science Emphasis), Berkeley AI Research, Launchpad ML Club (Head of Education)

EXPERIENCE

Berkeley AI Research | PALLAS Group **Sep. 2023 – Present**
Paid Researcher Berkeley, CA

- Building LLMCompiler, an open-source parallel function calling framework, with Professor Kurt Keutzer and research scientist Amir Gholami.
- Utilizing LLMCompiler to call web agents in parallel for efficient interaction with internet environments like WebShop and WebArena.
- Led experimentation on HotPotQA and Movie Recommendation benchmarks to compare latency against different branching factors using a custom WikiSearch API.

Insitro | Core Imaging Team **May 2023 – Aug. 2023**
Research Scientist Intern San Francisco, CA

- Developed IM2MEA, a system that aligns 6k+ microscopy images with 2.5k+ microelectrode array (MEA) recordings to retrieve activation potential signals for target validation of neurodegenerative diseases.
- Achieved cross-modal mapping by utilizing META AI's DINO segmentation model on a dataset of 5k image-MEA data points to impute MEA aggregate metrics and embedding centroids from image embeddings.
- Trained a transformer autoencoder on 2k MEA samples to reconstruct action potentials for downstream signal synthesis by encoding electrode-level and network-level MEA features.

Berkeley AI Research | Computer Vision Group **Sept. 2022 – Dec. 2022**
Researcher Berkeley, CA

- Assembled and installed ROS software on a FAIR LoCoBot NUC for robot navigation and object manipulation functionality with PhD student Ilija Radosavovic.
- Simulated policy training with scripts from ManiSkill2 benchmark and Isaac Gym on a GPU server.

NASA JPL | ML and Instrument Autonomy Team **May 2022 – Sep. 2022**
Research Scientist Intern Palo Alto, CA

- Headed research and trade study of 5 image compression algorithms, including the ICER Progressive Wavelet Image Compressor and JPEG 2000, with 500+ preprocessed images across Mars, Miranda, and Ceres mission data.
- Designed compression algorithm analysis pipeline with Python wrappers for C codebases used in deep space exploration, utilizing structural similarity indexes and heat maps as benchmark metrics.
- Enabled and streamlined optimization of compression algorithms through pipeline's analytics for future missions within JPL's Machine Learning and Instrument Autonomy Group and NASA at large.

PAPERS

An LLM Compiler for Parallel Function Calling https://arxiv.org/abs/2312.04511 **ArXiv Preprint**
Sehoon Kim*, Suhong Moon*, Ryan Tabrizi, Nicholas Lee, Michael W. Mahoney, Kurt Keutzer, Amir Gholami
LLMCompiler: A open-source framework for efficiently executing multiple tasks in parallel. LLMCompiler manages dependencies and executes user-provided APIs, achieving up to 1.3x speedup over OpenAI's Parallel Function Calling, as well as up to 9% higher accuracy compared to ReAct.

Convex Polytopes Are All You Need https://ryantabrizi.com/files/ConvexPolytopes.pdf **Technical Report**
Alberto Hojel*, Ryan Tabrizi*, Heather Ding*
A technical report on recent adversarial attacks (L-BFGS, FGSM) against computer vision models and corresponding defense mechanisms. We analyze how such attacks can be provably defended against by using convex polytopes and strong duality.

PROJECTS

Phodexr | CLIP, BERT, ResNet, Hugging Face, WandB

- Recreated Apple's camera roll semantic search with OpenAI's CLIP for lightweight album indexing from user queries.
- Deployed a pretrained DistilBERT tokenizer and trained ResNet50 on COCO Captions (~350k image-caption pairs) for image-text embedding similarity search.

SKILLS

- **Programming Languages:** Python, Java, C, LaTeX, Scheme (Lisp), RISC-V
- **Libraries/Frameworks:** PyTorch, NumPy, Hugging Face, LangChain, TensorFlow, Keras, scikit-learn, Pandas
- **Developer Tools:** Git, AWS, Linux, WandB, Streamlit, Gradio, Isaac Gym, STM32Cube